RAK12021 WisBlock RGB Sensor Module Datasheet

Overview

Description

RAK12021 is a WisBlock RGB Sensor that extends the WisBlock system which is based on TCS37725FN from AMS. The Red, Green, Blue, and Clear (RGBC) light sensing can be obtained via I2C interface. An external IR LED is also added for Proximity Detection.

Features

- RGB sensor module
- Color light sensing with IR-Blocking filter
- Proximity detection
- Maskable light and proximity interrupt
- Low power (2.5 uA sleep current)
- I2C interface
- 3.3 V power supply
- Current Consumption: 2.5 uA - 235 uA
- Chipset: AMS TCS37725FN
- **Module size**: 10 × 10 mm

Specifications

Overview

Mounting

Figure 1 shows the mounting mechanism of the RAK12021 module on a WisBlock Base board. The RAK12021 module can be mounted on the slots: **A, C, D, E, & F**.

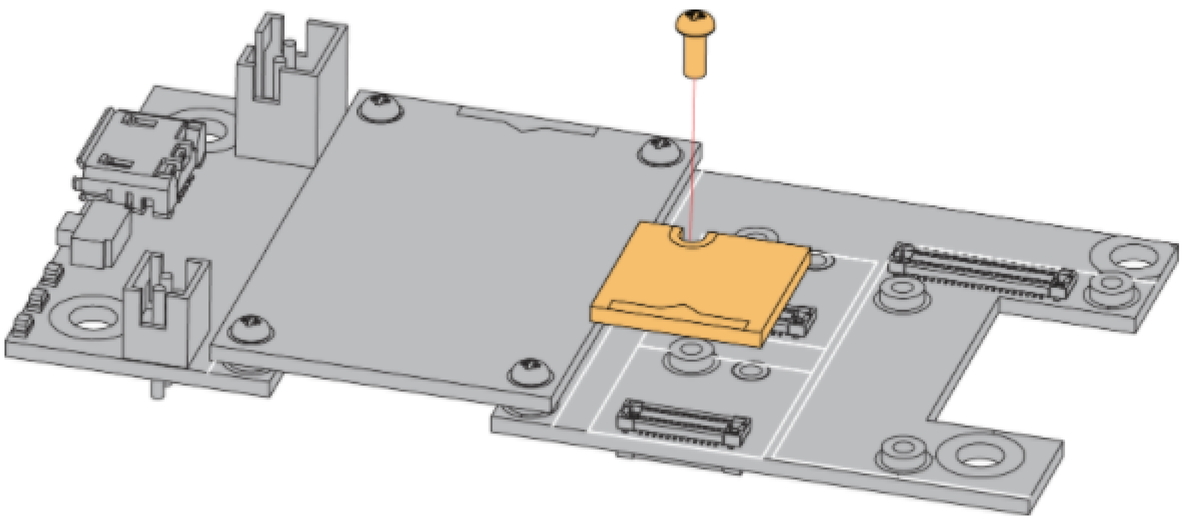

Figure 1: RAK12021 mounting mechanism

Hardware

The hardware specification is categorized into five (5) parts. It shows the chipset of the module and discusses the pinouts and their corresponding functions and diagrams. It also covers the electrical and mechanical parameters that include the tabular data of the functionalities and standard values of the RAK12021 WisBlock RGB Sensor Module.

Chipset

Vendor	Part number
AMS	TCS37725FN

Pin Definition

The RAK12021 WisBlock RGB Sensor Module comprises a standard WisBlock connector. The WisBlock connector allows the RAK12021 module to be mounted to a WisBlock Base board. The pin order of the connector and the pinout definition is shown in **Figure 2**.

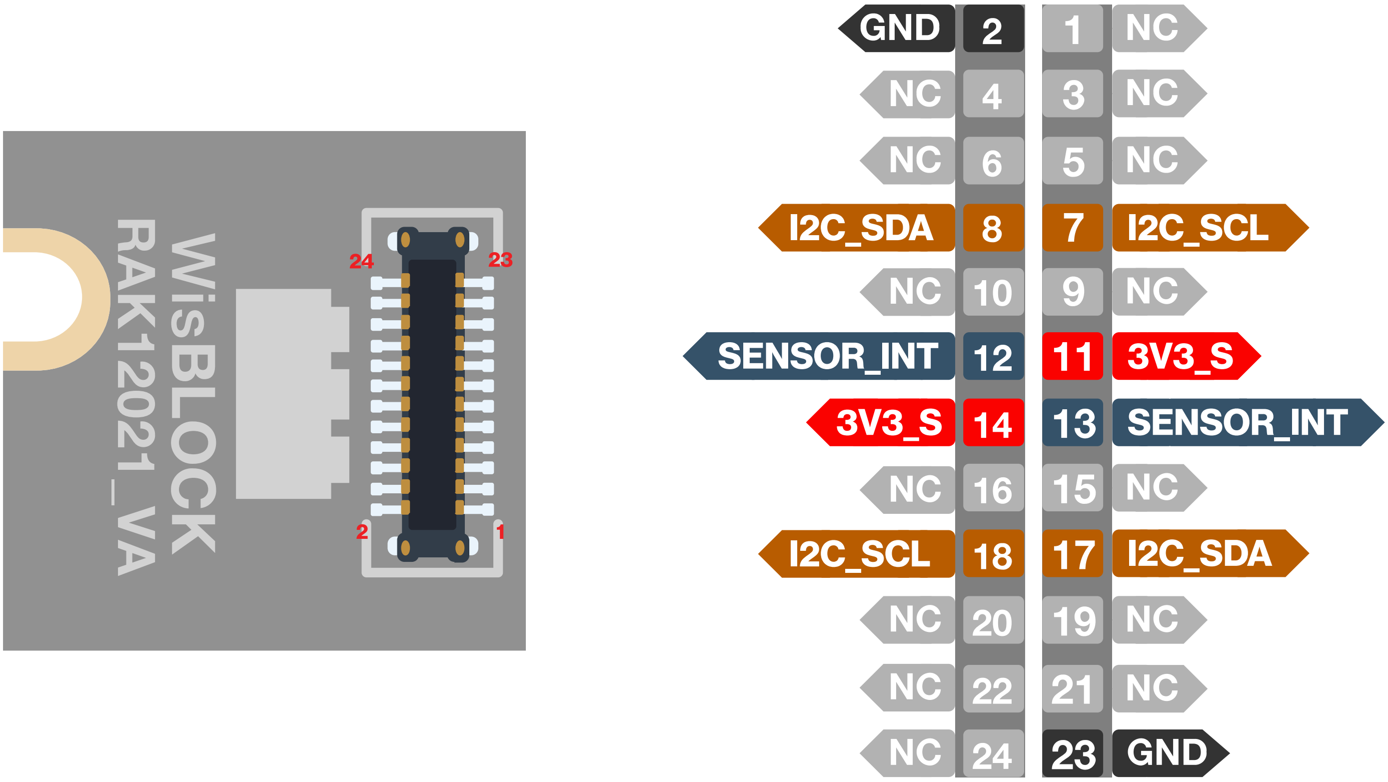

Figure 2: RAK12021 pinout

If a 24-pin WisBlock Sensor connector is used, the IO used for the output pulse depends on what slot the module is plugged in. The following table shows the default IO used for different slots:

SLOT A	SLOT C	SLOT D	SLOT E	SLOT F
WB_IO1	WB_IO3	WB_IO5	WB_IO4	WB_IO6

INT (Interrupt Pin)
WB_IO2

NOTE

- Only the **I2C** related pin, **SENSOR_INT**, **3V3_S**, and **GND** are connected to this module.
- **3V3_S** voltage output from the WisBlock Base that powers the RAK12021 module can be controlled by the WisBlock Core via WB_IO2 (WisBlock IO2 pin). This makes the

> module ideal for low-power IoT projects since the WisBlock Core can totally disconnect the power of the RAK12021 module.

Electrical Characteristics

Absolute Maximum Ratings

Parameter	Min.	Max.	Unit
3V3_S	-0.5	3.8	V
Output terminal current (except LDR)	-1	20	uA
ESD tolerance, human body model	-	±2000	V

Power Supply Ratings

Symbol	Description	Condition	Min.	Nom.	Max.	Unit
3V3_S	Supply voltage	Supply voltage	2.7	3.3	3.6	V
I_{DD1}	Operation mode current	Active - LDR pulses off	-	235	-	uA
I_{DD2}	Operation mode current	Wait state	-	65	-	uA
I_{DD3}	Operation mode current	Sleep state - no I2C activity	-	2.5	10	uA

Mechanical Characteristics

Board Dimensions

Figure 3 shows the dimensions and the mechanic drawing of the RAK12021 module.

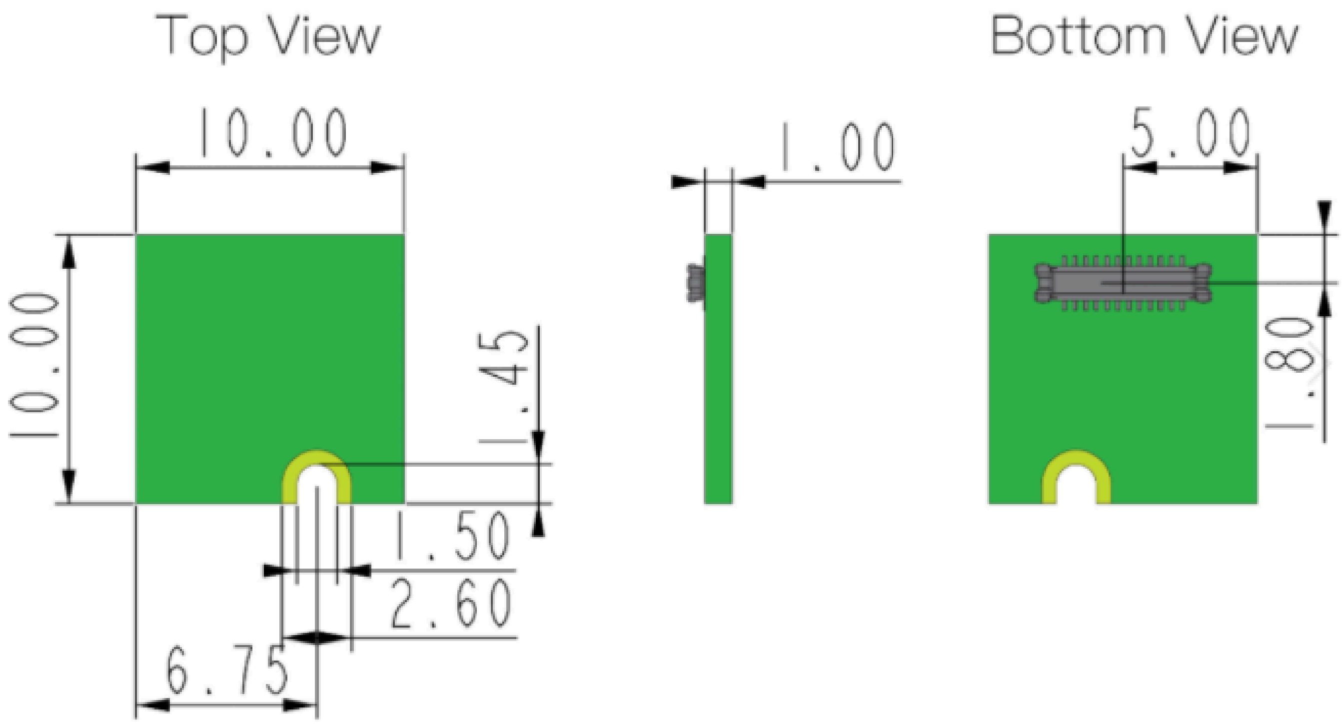

Figure 3: RAK12021 mechanical drawing

WisConnector PCB Layout

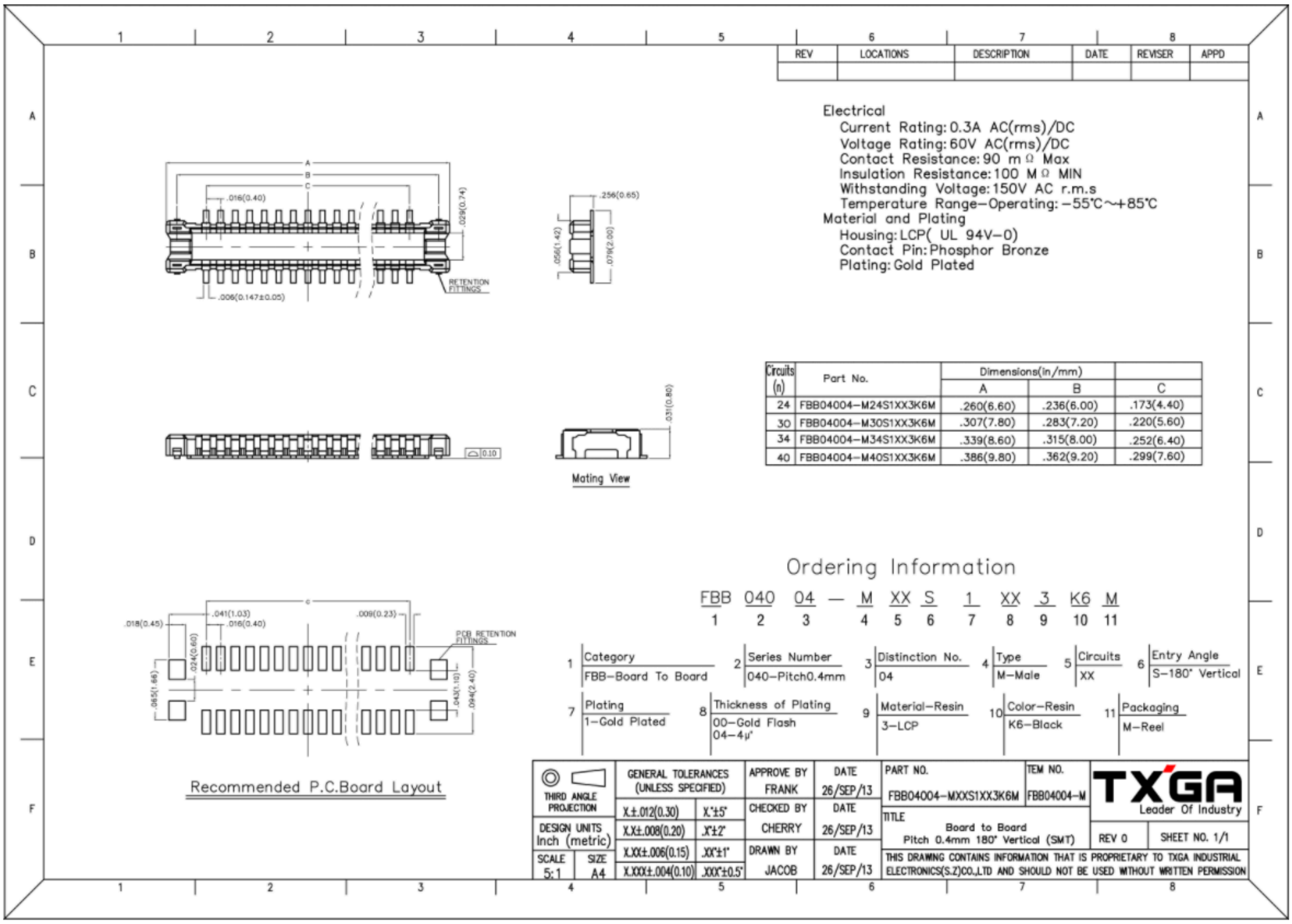

Figure 4: WisConnector PCB footprint and recommendations

Schematic Diagram

Figure 5 shows the schematic of the RAK12021 module.

D1 is the external IR LED for proximity detection.

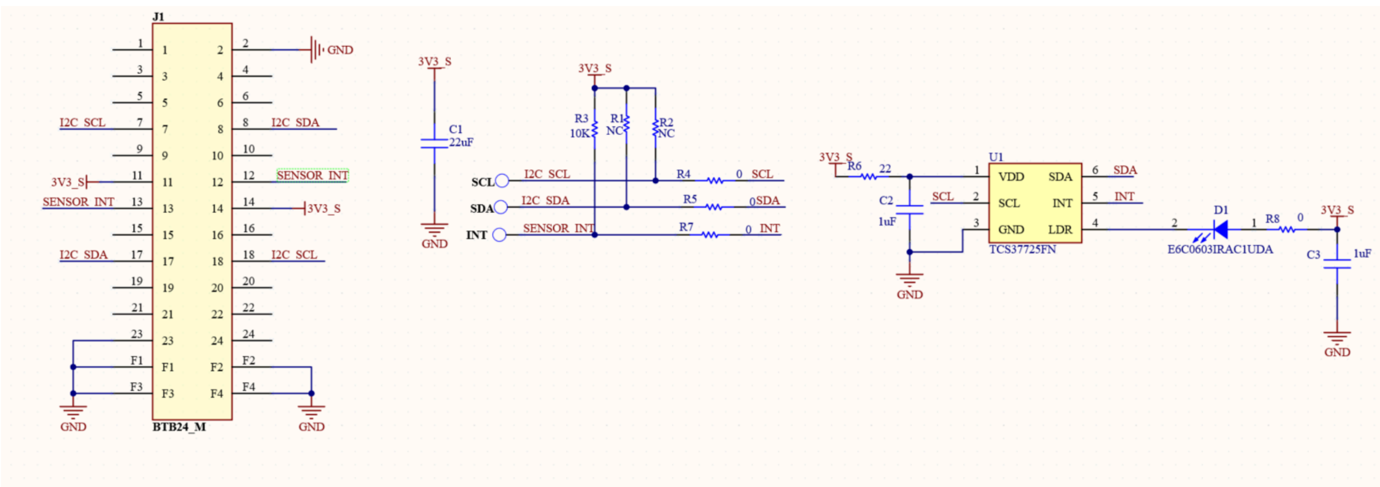

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Figure 5: RAK12021 schematics

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